WORKED SOLUTION

STEP II 2017, Question 3

Pure • Curve Sketching

STEP II 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q3

The key idea

The equation $\sin y = k$ never has just one solution: if y_0 is one, so is $\pi - y_0$, and so is anything differing from these by 2π. So every sketch here is a handful of copies of one basic curve, placed by the symmetries of sine. Part (i) shows the pattern with straight lines; in (ii) the calculus pins down the shape of the basic piece $y = \arcsin(\frac{1}{2}\sin x)$, and the symmetries do the rest. Part (iii) is (ii) again after writing $\cos y$ as a sine.

Solution

Part (i)

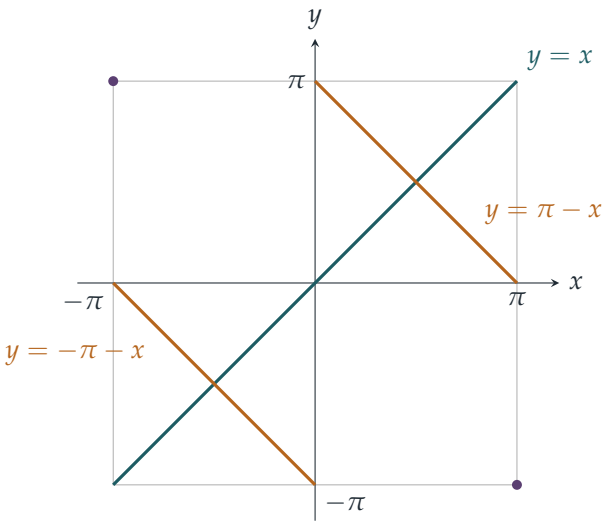
We have $\sin y = \sin x$ exactly when

$$y = x + 2k\pi \qquad \text{or} \qquad y = \pi - x + 2k\pi \qquad (k \in \mathbb{Z}).$$

Now we see which of these lines meet the square $-\pi \le x, y \le \pi$.

- $y = x$ runs along the whole diagonal, from $(-\pi, -\pi)$ to (π, π).
- $y = x \pm 2\pi$ need $y - x = \pm 2\pi$. In the square, $|y - x| \le 2\pi$ with equality only at the corners, so these give just the two isolated points $(-\pi, \pi)$ and $(\pi, -\pi)$.
- $x + y = \pi$ is the segment from $(0, \pi)$ to $(\pi, 0)$, and $x + y = -\pi$ (the case $k = -1$) is the segment from $(-\pi, 0)$ to $(0, -\pi)$.
- $x + y = 3\pi$ or -3π is impossible, since $|x + y| \le 2\pi$.

So the set is the diagonal $y = x$, the two segments of $x + y = \pm\pi$, and the two corner points $(-\pi, \pi)$ and $(\pi, -\pi)$.



The isolated corner points are easy to miss. They are genuine solutions, since $\sin(\pm\pi) = 0$.

Part (ii)

Differentiating $\sin y = \frac{1}{2}\sin x$ implicitly gives $y'\cos y = \frac{1}{2}\cos x$. For $0 \le y \le \frac{1}{2}\pi$ we have $\cos y \ge 0$, so

$$\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-\tfrac{1}{4}\sin^2 x} = \tfrac{1}{2}\sqrt{4-\sin^2 x},$$

which is never zero. Hence

$$y' = \frac{\cos x}{\sqrt{4-\sin^2 x}}.$$

Write $s = \sin x$. By the quotient rule, using $\dfrac{\mathrm{d}}{\mathrm{d}x}\sqrt{4-s^2} = \dfrac{-s\cos x}{\sqrt{4-s^2}}$,

$$y'' = \frac{-\sin x\sqrt{4-s^2} - \cos x\cdot\dfrac{-s\cos x}{\sqrt{4-s^2}}}{4-s^2} = \frac{-s(4-s^2)+s\cos^2 x}{(4-s^2)^{3/2}}$$
$$= \frac{s\left(-4+s^2+1-s^2\right)}{(4-s^2)^{3/2}} = -\frac{3\sin x}{(4-\sin^2 x)^{3/2}},$$

using $\cos^2 x = 1-s^2$. This is the required result.

The basic piece. In this quarter-square the curve is $y = f(x)$, where $f(x) = \arcsin(\frac{1}{2}\sin x)$. It starts at $(0,0)$ and ends at $(\frac{1}{2}\pi, \frac{1}{6}\pi)$, because $\arcsin\frac{1}{2} = \frac{1}{6}\pi$, so it never gets near the top edge $y = \frac{1}{2}\pi$. Its gradient is $\frac{1}{2}$ at the origin and falls to 0 at $x = \frac{1}{2}\pi$. Also $y'' < 0$ for $0 < x \le \frac{1}{2}\pi$ and $y'' = 0$ at $x = 0$, so the curve is concave and increasing, with a point of inflection at the origin.

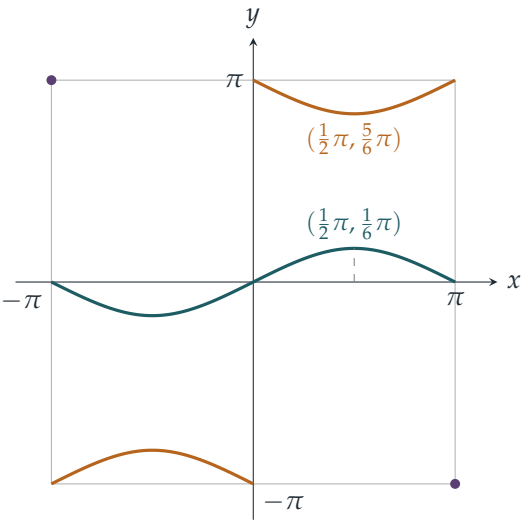
Ask yourself How do we get the whole square from this little arc without doing any more calculus?

We use symmetry. Since $\sin(\pi - x) = \sin x$ and f is odd, the graph of $y = f(x)$ on $[-\pi, \pi]$ is symmetric about $x = \frac{1}{2}\pi$ and has rotational symmetry about the origin. It is a gentle wave through $(-\pi, 0)$, $(0, 0)$ and $(\pi, 0)$, with maximum $(\frac{1}{2}\pi, \frac{1}{6}\pi)$ and minimum $(-\frac{1}{2}\pi, -\frac{1}{6}\pi)$. Its gradient is $\frac{1}{2}$ at the origin and $-\frac{1}{2}$ at $x = \pm\pi$.

As in (i), the full solution set is given by $y = f(x) + 2k\pi$ or $y = \pi - f(x) + 2k\pi$. Within the square this leaves:

- the wave $y = f(x)$ for all $-\pi \leq x \leq \pi$;
- $y = \pi - f(x)$, which lies in the square only where $f(x) \geq 0$, that is for $0 \leq x \leq \pi$. This is an arch hanging down from $(0, \pi)$ to (π, π), lowest at $(\frac{1}{2}\pi, \frac{5}{6}\pi)$;
- $y = -\pi - f(x)$, which lies in the square only where $f(x) \leq 0$, that is for $-\pi \leq x \leq 0$. This is an arch rising from $(-\pi, -\pi)$ to $(0, -\pi)$, highest at $(-\frac{1}{2}\pi, -\frac{5}{6}\pi)$;
- the isolated corner points $(-\pi, \pi)$ and $(\pi, -\pi)$, from $y = \pi - f(x)$ at $x = -\pi$ and $y = -\pi - f(x)$ at $x = \pi$.

The other values of k give $|y| > \pi$. The arches meet the lines $y = \pm\pi$ at gradient $\pm\frac{1}{2}$, and are the wave rotated through a half-turn and moved up or down. Compare (i): the diagonal has flattened into the wave and the two segments have bent into arches.



Part (iii)

Ask yourself How can we turn $\cos y$ into something we've already sketched?

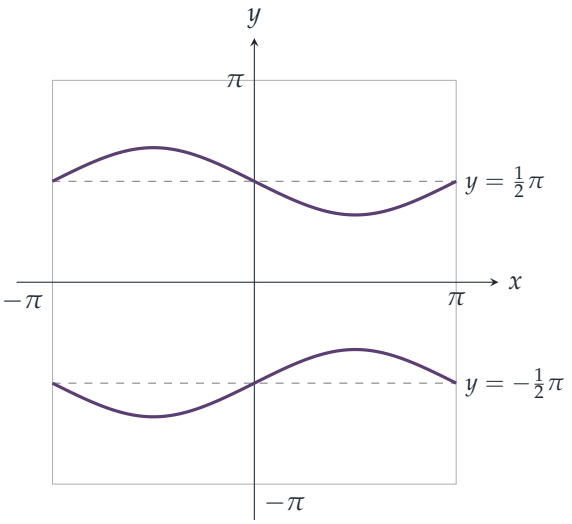
Write $\cos y = \sin(\frac{1}{2}\pi - y)$. Then $\cos y = \frac{1}{2}\sin x$ says that $u = \frac{1}{2}\pi - y$ satisfies $\sin u = \frac{1}{2}\sin x$, so $u = f(x) + 2k\pi$ or $u = \pi - f(x) + 2k\pi$. That is,

$$y = \tfrac{1}{2}\pi - f(x) - 2k\pi \qquad \text{or} \qquad y = f(x) - \tfrac{1}{2}\pi - 2k\pi.$$

Since $|f(x)| \le \frac{1}{6}\pi$, only $k = 0$ gives values in $[-\pi, \pi]$, and then it does so for every x. So the set consists of exactly two complete waves:

- $y = f(x) - \frac{1}{2}\pi$: the wave from (ii) moved down by $\frac{1}{2}\pi$, oscillating between $-\frac{2}{3}\pi$ and $-\frac{1}{3}\pi$;
- $y = \frac{1}{2}\pi - f(x)$: its mirror image in the x-axis (as it must be, since $\cos y$ is even in y). It has minimum $(\frac{1}{2}\pi, \frac{1}{3}\pi)$ and maximum $(-\frac{1}{2}\pi, \frac{2}{3}\pi)$.

There are no isolated points this time.



Checks

- Sample points: $(\frac{1}{2}\pi, \frac{1}{6}\pi)$ gives $\sin\frac{1}{6}\pi = \frac{1}{2} = \frac{1}{2}\sin\frac{1}{2}\pi$, and $(\frac{1}{2}\pi, \frac{5}{6}\pi)$ works too. In (iii), $(\frac{1}{2}\pi, \frac{1}{3}\pi)$ gives $\cos\frac{1}{3}\pi = \frac{1}{2}$.
- At $x = 0$ the formula gives $y' = \frac{1}{2}$, matching the small-angle approximation $y \approx \frac{1}{2}x$. At $x = \frac{1}{2}\pi$ it gives $y' = 0$, as it must at the maximum.
- A computer check confirmed the formulas for y' and y'' by finite differences at random points of $(0, \frac{1}{2}\pi)$. It also found every solution $y \in [-\pi, \pi]$ for many random x, and they matched the branches above: two in (ii) for $x \neq 0, \pm\pi$, and always two in (iii).

Common slips

- Missing the corner points $(-\pi, \pi)$ and $(\pi, -\pi)$ in (i) and (ii).
- Drawing the arches in (ii) over the whole of $[-\pi, \pi]$: $y = \pi - f(x)$ leaves the square as soon as $f(x) < 0$.
- Taking $\cos y = -\frac{1}{2}\sqrt{4 - \sin^2 x}$: the sign is fixed by $0 \leq y \leq \frac{1}{2}\pi$.

Beyond STEP

A set like $\{(x, y) : F(x, y) = 0\}$ is a *level set*. The implicit function theorem says it is locally a smooth graph wherever $\partial F/\partial y \neq 0$. Here that fails exactly where $\cos y = 0$, and since that never happens when $|\sin y| \leq \frac{1}{2}$, the curves in (ii) are smooth graphs over x. In (i), the crossing points are where both partial derivatives of $\sin y - \sin x$ vanish, and that is where the level set fails to be a curve. Bartle and Sherbert, *Introduction to Real Analysis*, builds the single-variable groundwork; Abbott, *Understanding Analysis*, is a lively companion.

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